

PC-DOCTOR

A Rule-Based Expert System for Windows PC Troubleshooting

Logic Programming and Artificial Cognitive Systems

CM3321

Expert System Assignment

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Code repository: <https://github.com/Heshan2691/pc-doctor-expert-system.git>

Table of Content

1. Introduction.....	4
1.1 Domain and problem	4
1.2 Objectives	4
1.3 Scope.....	4
1.4 Source of knowledge	4
2. Explanation of the System	4
2.1 Architecture.....	4
2.2 How a consultation works.....	5
2.3 Development tools	6
3. User Interface	6
3.1 Main menu	7
3.2 Consultation dialogue	7
3.3 Result and explanation	8
4. Knowledge Base	11
4.1 Knowledge acquisition.....	11
4.2 Knowledge representation.....	11
4.3 Rule summary.....	12
4.4 Rule ordering	13
5. Inference Engine	14
5.1 Backward chaining.....	14
5.2 Working memory.....	14
5.3 Conflict resolution	14
5.4 Explanation facility	14
5.5 When no rule matches.....	14
6. Test Cases	15
6.1 Test strategy	15
6.2 Automated tests.....	15
6.3 Interactive tests.....	16
6.4 Summary.....	19
7. How to Run the System Locally	19
7.1 Requirements	19
7.2 Steps.....	19
7.3 Project files	20
8. References	20
Annex A – Complete Rule Base (34 rules)	22
Annex B – Decision-Making Code	24

B.1 knowledge_base.pl 24

B.2 engine.pl..... 29

Annex C – Automated Test Results (40 tests)..... 32

1. Introduction

1.1 Domain and problem

Windows users regularly run into the same kinds of problems: no sound, no Wi-Fi, a printer that will not print, blue screen errors, a blank screen, Bluetooth devices that will not connect, a slow PC, or Windows Update failures. Microsoft publishes step-by-step troubleshooting articles for each of these problems, but a non-technical user has to read long pages and decide which of many steps applies to their situation.

PC-Doctor captures this troubleshooting knowledge in an expert system. It asks short yes/no questions, identifies the most likely cause, recommends the fix that Microsoft gives for that case, and explains how it reached its conclusion.

1.2 Objectives

- Diagnose common Windows problems in 8 problem areas using at least 20 rules (34 rules implemented).
- Take every rule from a documented, authoritative source and cite it.
- Ask only the questions that are needed and never repeat a question.
- Explain the reasoning to the user (WHY a question is asked and HOW a conclusion was reached).
- Provide a simple interface that runs locally with one command.

1.3 Scope

#	Problem area	Rules	Source
1	Sound / audio problems	R01, R02, R03, R04	S1
2	Wi-Fi / internet problems	R05, R06, R07, R08, R09	S2
3	Printer problems	R10, R11, R12, R13	S3
4	Blue screen (stop code) errors	R14, R15, R16, R17, R18	S4 / S4 / S9
5	Blank or black screen	R19, R20, R21, R22	S5
6	Bluetooth problems	R23, R24, R25, R26	S6
7	Slow PC performance	R27, R28, R29, R30	S7 / S7 / S9
8	Windows Update problems	R31, R32, R33, R34	S8 / S8 / S9

1.4 Source of knowledge

No human expert was consulted. All knowledge and rules were taken from nine official Microsoft Support articles (S1–S9, listed in Section 8). Each rule records the article and the section heading it was derived from (Annex A). Where an article states a condition for a step (for example, "If you added new hardware to your PC before the error"), that condition became the IF part of the rule. Where an article lists a step without an explicit condition, the IF part was formed from the problem that step addresses; these rules are marked "Listed troubleshooting step" in the rule table.

2. Explanation of the System

2.1 Architecture

PC-Doctor follows the classic expert system architecture: a user interface, an inference engine with an explanation facility, a knowledge base, and a working memory. Figure 1 shows the components and how they interact.

Figure 1: PC-Doctor System Architecture

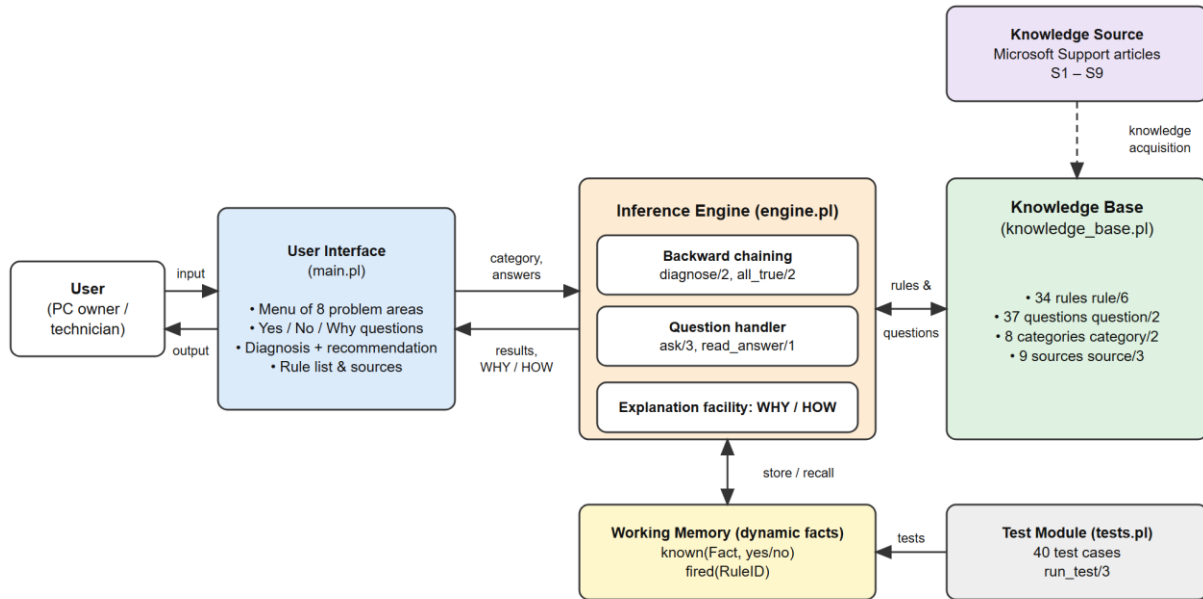


Figure 1: PC-Doctor system architecture

Component	File	Responsibility
User interface	main.pl	Shows the menu, reads the user's choice, starts a consultation, lists rules and sources.
Inference engine	engine.pl	Backward chaining over the rules of the chosen category; asks questions only when needed.
Explanation facility	engine.pl	WHY: shows the rule being tested. HOW: shows the rule that fired, the answers and the source.
Working memory	engine.pl (dynamic facts)	known(Fact, yes/no) stores answers; fired(RuleID) stores the conclusion.
Knowledge base	knowledge_base.pl	34 rules, 37 questions, 8 categories and 9 sources.
Test module	tests.pl	40 automated test cases that check every rule and special cases.

2.2 How a consultation works

1. The user starts the system (swipl main.pl) and selects a problem area from the menu.
2. The working memory is cleared.
3. The inference engine takes the rules of that problem area from top to bottom.
4. For each condition of a rule, the engine uses the stored answer if the question was already asked; otherwise it asks the user and stores the answer.
5. If all conditions of a rule are "yes", the rule fires and the diagnosis, recommendation and source are shown. If a condition is "no", the engine moves to the next rule.
6. If no rule fires, the system reports that it cannot make a diagnosis. The user can then ask HOW, or return to the menu.

Figure 2 shows this process as a flowchart.

Figure 2: Inference Process (Backward Chaining)

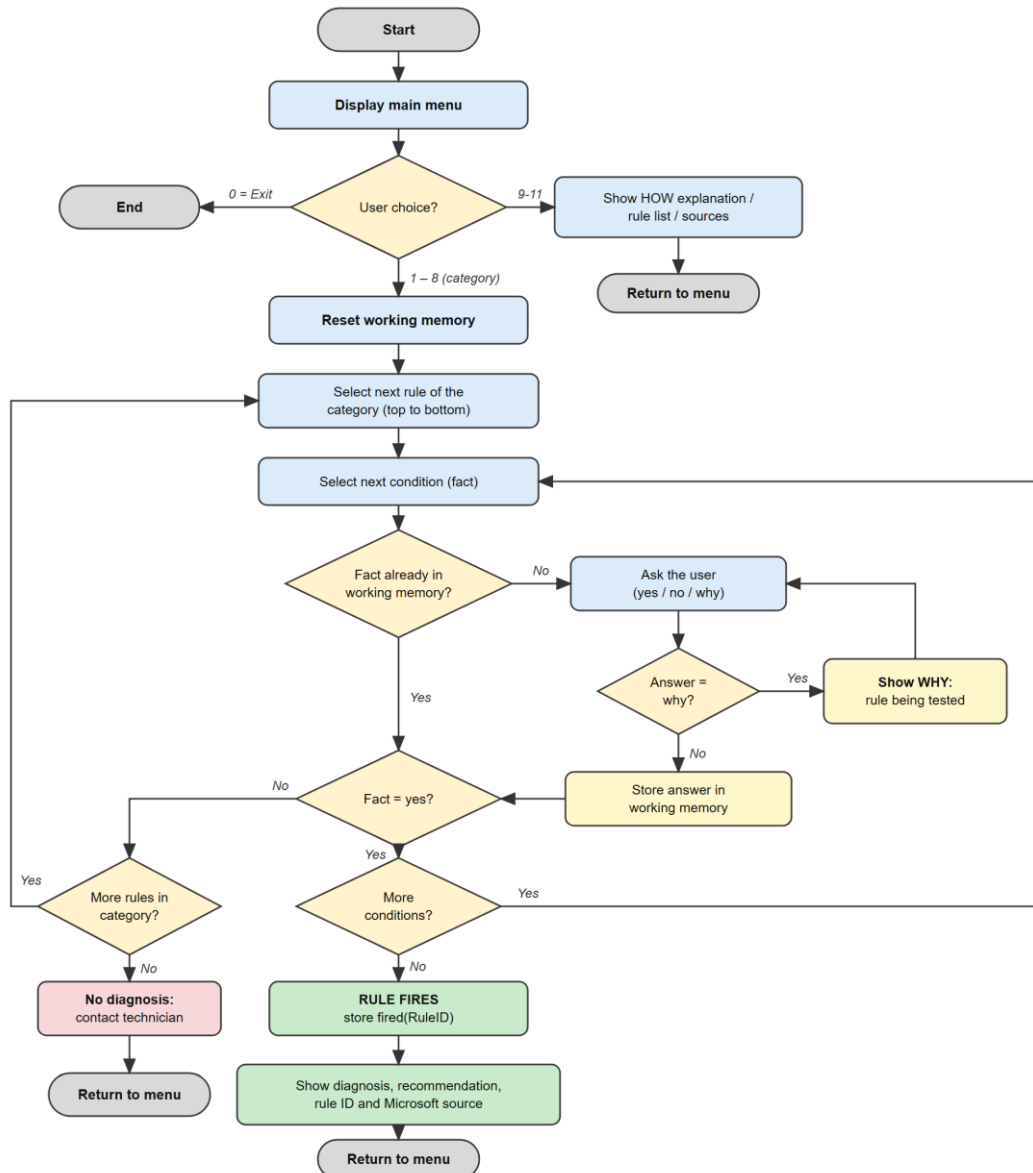


Figure 2: Inference process (backward chaining)

2.3 Development tools

Item	Choice
Language / shell	SWI-Prolog 10.0.2 (Prolog's built-in unification and backtracking give backward chaining directly)
Editor	Visual Studio Code
Interface type	Menu-driven command line interface (CLI)
Platform	Windows, macOS or Linux (any system that runs SWI-Prolog)

3. User Interface

PC-Doctor has a menu-driven command line interface. The user only types menu numbers and yes/no answers; no Prolog knowledge is needed.

3.1 Main menu

When the system starts it shows a banner with the number of loaded rules and a numbered menu (Screenshot 1).

Option	Function
1 – 8	Start a consultation for a problem area (sound, Wi-Fi, printer, blue screen, blank screen, Bluetooth, slow PC, Windows Update)
9	Explain last diagnosis (HOW)
10	Show all rules in IF ... THEN form with their source
11	Show the knowledge sources (Microsoft Support URLs)
0	Exit

```
what problem does your windows PC have?

1. Sound / audio problems
2. Wi-Fi / internet problems
3. Printer problems
4. Blue screen (stop code) errors
5. Blank or black screen
6. Bluetooth problems
7. Slow PC performance
8. Windows Update problems

9. Explain last diagnosis (HOW)
10. Show all rules
11. Show knowledge sources
0. Exit

Enter your choice: █
```

Screenshot 1: main menu after running `swipl main.pl`

3.2 Consultation dialogue

The system asks one yes/no question at a time. The user may answer yes/y, no/n, or why/w. Any other input is rejected with the message Please type yes, no or why. Typing why shows which rule is being tested, its conditions and which of them are already known, and then repeats the question (Screenshot 2).

```
Enter your choice: 3

=== Printer problems ===
(Answer yes or no. Type why to see why a question is asked.)

Is your printer not printing or not responding? (yes/no/why): yes

Does the printer show an "Offline" status? (yes/no/why): why

[WHY] I am testing rule R13:
  IF
    - Is your printer not printing or not responding? [yes]
    - Does the printer show an "Offline" status? [not asked yet]
  THEN Printer is set to Offline
  (Source: Fix printer connection and printing problems in Windows)

Does the printer show an "Offline" status? (yes/no/why): no

Are print jobs stuck in the print queue? (yes/no/why): yes

-----
DIAGNOSIS      : Print spooler problem
RECOMMENDATION : Clear and reset the print spooler service.
RULE / SOURCE  : R12 - Microsoft Support: Fix printer connection and printing problems in Windows
-----

Press Enter to return to the menu...[]
```

Screenshot 2: printer consultation with a WHY explanation and the diagnosis (TC-I01)

3.3 Result and explanation

When a rule fires, the result box shows the diagnosis, the recommended action, and the rule ID and Microsoft source. Choosing option 9 (HOW) then lists the rule's conditions with the user's answers, the source URL and the rules that were ruled out by "no" answers (Screenshot 3).

```
Enter your choice: 9

[HOW] The conclusion was reached by rule R12:
  IF
    - Is your printer not printing or not responding? [yes]
    - Are print jobs stuck in the print queue? [yes]
  THEN Print spooler problem

  Source S3: Fix printer connection and printing problems in Windows
  https://support.microsoft.com/en-us/windows/fix-printer-connection-and-printing-problems-in-windows-fb830bff-7702-6349-33cd-9443fe987f73

  Rules ruled out by your 'no' answers: R13

Press Enter to return to the menu...[]
```

Screenshot 3: HOW explanation after the printer consultation (TC-I02)

Options 10 and 11 list the whole knowledge base and its sources, so the user can inspect every rule (Screenshot 4).

```

--- Sound / audio problems ---
R01: IF no_sound AND wrong_output_device
    THEN Audio output device is misconfigured [S1]
R02: IF no_sound AND audio_hardware_muted_or_unplugged
    THEN Audio hardware connection or volume issue [S1]
R03: IF no_sound AND after_update
    THEN Audio driver is incompatible after a Windows update [S1]
R04: IF no_sound AND audio_driver_problem
    THEN Outdated or corrupt audio driver [S1]

--- Wi-Fi / internet problems ---
R05: IF no_wifi AND airplane_mode_on
    THEN Wireless radios are disabled by Airplane mode [S2]
R09: IF no_wifi AND after_update
    THEN Network adapter driver problem after a recent update [S2]
R06: IF no_wifi AND saved_network_fails
    THEN Saved Wi-Fi network profile problem [S2]
R07: IF no_internet AND other_devices_fail
    THEN Router or modem problem (not the PC) [S2]
R08: IF no_internet AND other_devices_ok
    THEN Network configuration problem on this PC [S2]

--- Printer problems ---
R13: IF printer_not_working AND printer_offline
    THEN Printer is set to Offline [S3]
R12: IF printer_not_working AND jobs_stuck_in_queue
    THEN Print spooler problem [S3]
R11: IF printer_not_working AND printer_connection_problem
    THEN Printer connection problem [S3]
R10: IF printer_not_working
    THEN Printer is in a stuck state [S3]

--- Blue screen (stop code) errors ---
R14: IF blue_screen AND new_hardware_added
    THEN Newly added hardware is causing the stop error [S4]
R15: IF blue_screen AND cannot_restart
    THEN Windows cannot start normally [S4]

```

```

    THEN Windows cannot start normally [S4]
R16: IF blue_screen AND device_manager_warning
    THEN Faulty device driver [S4]
R17: IF blue_screen AND low_disk_space
    THEN Insufficient free disk space [S4]
R18: IF blue_screen AND steps_failed
    THEN Persistent system problem [S4]

--- Blank or black screen ---
R20: IF blank_screen AND cursor_visible
    THEN Windows Explorer has stalled [S5]
R19: IF blank_screen AND external_display
    THEN Display connection problem [S5]
R21: IF blank_screen AND after_update
    THEN Graphics driver problem after an update [S5]
R22: IF blank_screen AND began_recently
    THEN A recent system change caused the problem [S5]

--- Bluetooth problems ---
R23: IF bluetooth_problem AND bluetooth_off
    THEN Bluetooth is disabled on the PC [S6]
R24: IF bluetooth_problem AND bt_device_not_ready
    THEN Bluetooth device is not ready [S6]
R26: IF bluetooth_problem AND bt_missing_but_in_device_manager
    THEN Bluetooth adapter is not loaded properly [S6]
R25: IF bluetooth_problem AND after_update
    THEN Outdated or incompatible Bluetooth driver [S6]

--- Slow PC performance ---
R30: IF slow_pc AND suspicious_activity
    THEN Possible malware infection [S7]
R27: IF slow_pc AND low_disk_space
    THEN Low disk space is reducing performance [S7]
R28: IF slow_pc AND slow_startup
    THEN Too many startup apps [S7]
R29: IF slow_pc AND many_apps_open
    THEN Too many open apps are using resources [S7]

```

```

--- Slow PC performance ---
R30: IF slow_pc AND suspicious_activity
    THEN Possible malware infection [S7]
R27: IF slow_pc AND low_disk_space
    THEN Low disk space is reducing performance [S7]
R28: IF slow_pc AND slow_startup
    THEN Too many startup apps [S7]
R29: IF slow_pc AND many_apps_open
    THEN Too many open apps are using resources [S7]

--- Windows Update problems ---
R31: IF update_fails AND no_stable_internet
    THEN Update cannot download or install [S8]
R32: IF update_fails AND low_disk_space
    THEN Not enough free space for the update [S8]
R33: IF update_fails AND wrong_date_time
    THEN Incorrect date and time settings [S8]
R34: IF update_fails AND update_error_persists
    THEN Corrupt update files in the Windows Update cache [S8]

Press Enter to return to the menu...

```

Screenshot 4: list of all rules (option 10)

4. Knowledge Base

4.1 Knowledge acquisition

1. Nine Microsoft Support troubleshooting articles covering eight common Windows problem areas were selected (S1–S9).
2. Each article was read and every troubleshooting step with its stated condition was recorded.
3. Each step became one IF-THEN rule: the IF part is the symptom plus the condition; the THEN part is the cause and the fix.
4. The article and the exact section heading were recorded for every rule (Annex A), and copies of the pages were saved on 28 September 2026.

4.2 Knowledge representation

The knowledge base (knowledge_base.pl) contains only facts; it has no reasoning code. Four kinds of facts are used:

Predicate	Meaning	Example
source/3	A Microsoft Support article	source(s4, 'Resolving Blue Screen errors in Windows', URL)
category/2	A problem area shown in the menu	category(printer, 'Printer problems')
question/2	The yes/no question for a fact	question(cursor_visible, 'Can you see the mouse cursor...?')
rule/6	An IF-THEN rule	rule(ID, Category, Conditions, Diagnosis, Action, Source)

Example rule:

```

% R14 - S4: "Remove any new hardware"
rule(r14, blue_screen, [blue_screen, new_hardware_added],

```

```
'Newly added hardware is causing the stop error',
'Remove the new hardware and restart the PC.',
s4).
```

This reads: **IF** the user has a blue screen error **AND** new hardware was added before the error, **THEN** the newly added hardware is causing the stop error; the recommended action is to remove it and restart. Source: S4.

4.3 Rule summary

The knowledge base contains 34 rules, 37 questions, 8 categories and 9 sources. The complete rule base with the source section of every rule is given in Annex A; the Prolog code is given in Annex B.

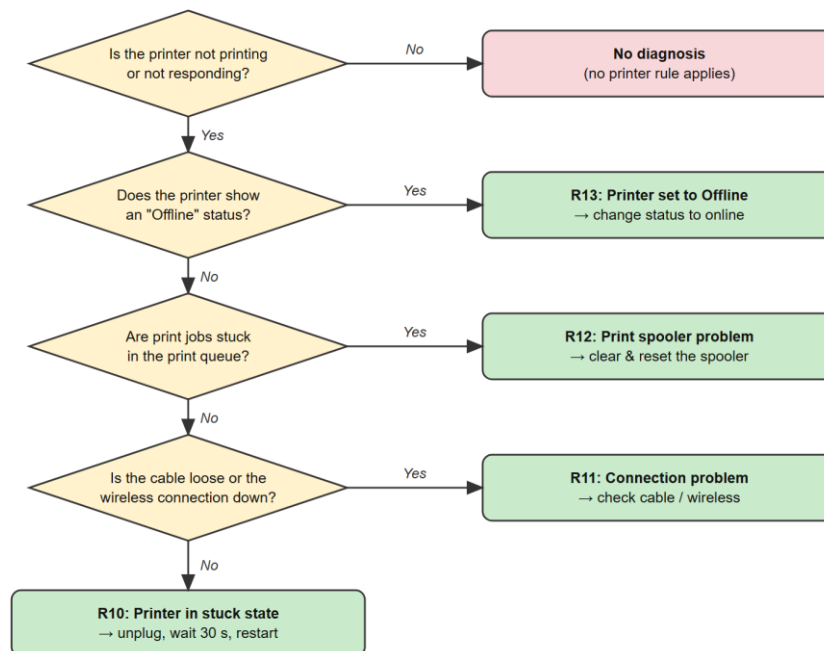
Rule	Area	IF (conditions)	THEN (diagnosis)
R01	Sound	no_sound AND wrong_output_device	Audio output device is misconfigured
R02	Sound	no_sound AND audio_hardware_muted_or_unplugged	Audio hardware connection or volume issue
R03	Sound	no_sound AND after_update	Audio driver is incompatible after a Windows update
R04	Sound	no_sound AND audio_driver_problem	Outdated or corrupt audio driver
R05	Wi-Fi	no_wifi AND airplane_mode_on	Wireless radios are disabled by Airplane mode
R06	Wi-Fi	no_wifi AND saved_network_fails	Saved Wi-Fi network profile problem
R07	Wi-Fi	no_internet AND other_devices_fail	Router or modem problem (not the PC)
R08	Wi-Fi	no_internet AND other_devices_ok	Network configuration problem on this PC
R09	Wi-Fi	no_wifi AND after_update	Network adapter driver problem after a recent update
R10	Printer	printer_not_working	Printer is in a stuck state
R11	Printer	printer_not_working AND printer_connection_problem	Printer connection problem
R12	Printer	printer_not_working AND jobs_stuck_in_queue	Print spooler problem
R13	Printer	printer_not_working AND printer_offline	Printer is set to Offline
R14	Blue screen (stop code)	blue_screen AND new_hardware_added	Newly added hardware is causing the stop error
R15	Blue screen (stop code)	blue_screen AND cannot_restart	Windows cannot start normally
R16	Blue screen (stop code)	blue_screen AND device_manager_warning	Faulty device driver
R17	Blue screen (stop code)	blue_screen AND low_disk_space	Insufficient free disk space
R18	Blue screen (stop code)	blue_screen AND steps_failed	Persistent system problem
R19	Blank or black screen	blank_screen AND external_display	Display connection problem
R20	Blank or black screen	blank_screen AND cursor_visible	Windows Explorer has stalled
R21	Blank or black screen	blank_screen AND after_update	Graphics driver problem after an update
R22	Blank or black screen	blank_screen AND began_recently	A recent system change caused the problem
R23	Bluetooth	bluetooth_problem AND bluetooth_off	Bluetooth is disabled on the PC
R24	Bluetooth	bluetooth_problem AND	Bluetooth device is not ready

Rule	Area	IF (conditions)	THEN (diagnosis)
		bt_device_not_ready	
R25	Bluetooth	bluetooth_problem AND after_update	Outdated or incompatible Bluetooth driver
R26	Bluetooth	bluetooth_problem AND bt_missing_but_in_device_manager	Bluetooth adapter is not loaded properly
R27	Slow PC performance	slow_pc AND low_disk_space	Low disk space is reducing performance
R28	Slow PC performance	slow_pc AND slow_startup	Too many startup apps
R29	Slow PC performance	slow_pc AND many_apps_open	Too many open apps are using resources
R30	Slow PC performance	slow_pc AND suspicious_activity	Possible malware infection
R31	Windows Update	update_fails AND no_stable_internet	Update cannot download or install
R32	Windows Update	update_fails AND low_disk_space	Not enough free space for the update
R33	Windows Update	update_fails AND wrong_date_time	Incorrect date and time settings
R34	Windows Update	update_fails AND update_error_persists	Corrupt update files in the Windows Update cache

4.4 Rule ordering

Inside each category, rules are ordered from most specific to least specific, because the inference engine uses the first rule whose conditions are all true. For example, R10 (restart the printer) needs only printer_not_working, so it is placed after R13, R12 and R11, which need an extra, more specific condition. Figure 3 shows the resulting question sequence for printer problems.

Figure 3: Example Decision Path - Printer Problems (Rules R10-R13)



Source S3: Microsoft Support - Fix printer connection and printing problems in Windows

Figure 3: Decision path for printer problems (rules R10–R13)

5. Inference Engine

5.1 Backward chaining

The inference engine (`engine.pl`) is goal-driven. To diagnose a category it tries to prove the conditions of each rule of that category in turn. Prolog's backtracking moves to the next rule automatically when a condition fails.

```
diagnose(Category, RuleID) :-
    rule(RuleID, Category, Conditions, _, _, _), % take the next rule
    all_true(Conditions, RuleID),               % prove every condition
    !,                                           % stop at the first rule that fires
    assertz(fired(RuleID)).

is_true(Fact, _) :- known(Fact, yes), !.        % answer already known: yes
is_true(Fact, _) :- known(Fact, no), !, fail.   % answer already known: no
is_true(Fact, RuleID) :- ask(Fact, RuleID, Answer), % otherwise ask the user
    assertz(known(Fact, Answer)),
    Answer == yes.
```

5.2 Working memory

Answers are stored as dynamic facts `known(Fact, yes)` or `known(Fact, no)`, and the rule that gives the result is stored as `fired(RuleID)`. The working memory is cleared at the start of each consultation. Because every answer is stored, a question used by several rules is asked only once. For example, the Wi-Fi question is used by rules R05, R09 and R06 but is asked a single time (test TC-I06).

5.3 Conflict resolution

When more than one rule could match, the first rule in the knowledge base order fires (Section 4.4). Test T38 checks this: with offline, jobs stuck and printer not working all answered yes, rule R13 fires.

5.4 Explanation facility

WHY – typed during a question, it shows the rule currently being tested, each of its conditions and whether it is already known. **HOW** – after a diagnosis, it shows the rule that fired, the user's answers that made it true, the source URL, and the rules that were ruled out. Example HOW output from the system:

```
[HOW] The conclusion was reached by rule R12:
IF
  - Is your printer not printing or not responding? [yes]
  - Are print jobs stuck in the print queue? [yes]
THEN Print spooler problem

Source S3: Fix printer connection and printing problems in Windows
https://support.microsoft.com/en-us/windows/fix-printer-connection-and-printing-...

Rules ruled out by your 'no' answers: R13
```

5.5 When no rule matches

If none of the rules of the chosen category fires, the system does not guess. It reports that no diagnosis can be made and advises the user to contact a technician or Microsoft Support (tests T35–T37 and TC-I05).

6. Test Cases

6.1 Test strategy

- **Automated tests** (tests.pl, 40 cases): one test per rule (T01–T34), three "no diagnosis" cases (T35–T37) and three conflict resolution cases (T38–T40). Unlisted questions are answered "no".
- **Interactive tests** (main.pl, 11 cases): the real user interface, including WHY, HOW, invalid input and menu options.

6.2 Automated tests

Run with `swipl -g run_all_tests -t halt tests.pl`. A selection is shown below; the full list is in **Annex C**.

Test	Category	Purpose	User answers	Expected	Actual	Result
T01	Sound / audio problems	Rule R01 fires when its conditions are true	YES: Is there no sound coming from your PC? YES: Is the wrong (or no) output device selected in Sound settings?	R01: Audio output device is misconfigured	R01: Audio output device is misconfigured	PASS
T10	Printer problems	Rule R10 fires when its conditions are true	YES: Is your printer not printing or not responding?	R10: Printer is in a stuck state	R10: Printer is in a stuck state	PASS
T14	Blue screen (stop code) errors	Rule R14 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Did you add new hardware before the error started?	R14: Newly added hardware is causing the stop error	R14: Newly added hardware is causing the stop error	PASS
T27	Slow PC performance	Rule R27 fires when its conditions are true	YES: Is your PC running slowly? YES: Is your hard drive almost full?	R27: Low disk space is reducing performance	R27: Low disk space is reducing performance	PASS
T35	Printer problems	Main symptom answered NO - system must not give a diagnosis	(all questions answered NO)	No diagnosis	No diagnosis	PASS
T37	Sound / audio problems	Main symptom YES but no detail matches - no diagnosis	YES: Is there no sound coming from your PC?	No diagnosis	No diagnosis	PASS
T38	Printer problems	Several details YES - most specific rule (first in order) must win	YES: Is your printer not printing or not responding? YES: Does the printer show an "Offline" status? YES: Are print jobs stuck in the print queue?	R13: Printer is set to Offline	R13: Printer is set to Offline	PASS
T40	Wi-Fi / internet problems	Several details YES - most specific rule (first in order) must win	YES: Are you unable to connect to Wi-Fi? YES: Is Airplane mode turned on? YES: Did the problem start after a Windows update?	R05: Wireless radios are disabled by Airplane mode	R05: Wireless radios are disabled by Airplane mode	PASS

Result: 40 of 40 automated tests passed (Screenshot 5).

```

PS D:\L3S2\Logic Programming\pc-doctor> swipl -g run_all_tests -t halt tests.pl
t14 blue_screen      r14      r14      PASS
t15 blue_screen      r15      r15      PASS
t16 blue_screen      r16      r16      PASS
t17 blue_screen      r17      r17      PASS
t18 blue_screen      r18      r18      PASS
t19 blank_screen     r19      r19      PASS
t20 blank_screen     r20      r20      PASS
t21 blank_screen     r21      r21      PASS
t22 blank_screen     r22      r22      PASS
t23 bluetooth        r23      r23      PASS
t24 bluetooth        r24      r24      PASS
t25 bluetooth        r25      r25      PASS
t26 bluetooth        r26      r26      PASS
t27 slow_pc          r27      r27      PASS
t28 slow_pc          r28      r28      PASS
t29 slow_pc          r29      r29      PASS
t30 slow_pc          r30      r30      PASS
t31 windows_update   r31      r31      PASS
t32 windows_update   r32      r32      PASS
t33 windows_update   r33      r33      PASS
t34 windows_update   r34      r34      PASS
t35 printer          none     none     PASS
t36 sound            none     none     PASS
t37 sound            none     none     PASS
t38 printer          r13     r13      PASS
t39 slow_pc          r30     r30      PASS
t40 wifi             r05     r05      PASS
-----
Passed 40 of 40 tests
PS D:\L3S2\Logic Programming\pc-doctor>

```

Screenshot 5: output of `swipl -g run_all_tests -t halt tests.pl` ending with 'Passed 40 of 40 tests'

6.3 Interactive tests

Test	Purpose	Menu	User input	Expected output	Actual
TC-I01	Diagnosis + WHY explanation (printer)	3	Printer not printing? yes Offline status? why -> then no Jobs stuck in queue? yes	WHY shows rule R13 is being tested; then DIAGNOSIS: Print spooler problem (R12, source S3)	As expected
TC-I02	HOW explanation	9 (after TC-I01)	-	Shows R12 fired, both YES answers, source S3 URL, and 'Rules ruled out by your no answers: R13'	As expected
TC-I03	Diagnosis (slow PC)	7	PC running slowly? yes Pop-ups / unknown programs? yes	DIAGNOSIS: Possible malware infection (R30, source S7)	As expected
TC-I04	Rule order / skipping a rule (blue screen)	4	Blue screen error? yes New hardware added? no Trouble restarting? yes	R14 is skipped because of the NO answer; DIAGNOSIS: Windows cannot start normally (R15, source S4)	As expected
TC-I05	No diagnosis when no rule matches (sound)	1	No sound? yes Wrong output device? no Speakers unplugged/muted? no Started after update? no Driver warning? no	'No rule matched your answers, so no diagnosis can be made.'	As expected
TC-	Working	2	Unable to connect to	Only 2 questions are	As

Test	Purpose	Menu	User input	Expected output	Actual
I06	memory - question not asked twice (Wi-Fi)		Wi-Fi? no Connected but no internet? no	asked (the Wi-Fi question is asked once, although 3 rules use it); then No diagnosis	expected
TC-I07	Invalid answer to a question	3	Printer not printing? maybe	'Please type yes, no or why:' and the question waits for a valid answer	As expected
TC-I08	Invalid menu choice	abc	-	'Invalid choice. Please enter a number from the menu.'	As expected
TC-I09	List knowledge base	10	-	All 34 rules listed by category in IF ... THEN form with source ID	As expected
TC-I10	List knowledge sources	11	-	The 9 Microsoft Support pages (S1-S9) with URLs	As expected
TC-I11	Exit the system	0	-	'Thank you for using PC-Doctor. Goodbye!' and the program closes	As expected

Result: all 11 interactive tests gave the expected output. The actual system output for the special cases TC-I05, TC-I06 and TC-I07 is shown below.

TC-I05 – no rule matches (sound):

```
Is there no sound coming from your PC? (yes/no/why): yes
Is the wrong (or no) output device selected in Sound settings? (yes/no/why): no
Are your speakers/headphones unplugged, powered off or muted? (yes/no/why): no
Did the problem start after a Windows update? (yes/no/why): no
Does your audio device show a warning in Device Manager? (yes/no/why): no
-----
No rule matched your answers, so no diagnosis can be made.
Please contact a technician or Microsoft Support.
-----
```

The main symptom question (Is there no sound?) is used by all four sound rules but was asked only once.

TC-I06 – working memory (Wi-Fi):

```
Are you unable to connect to Wi-Fi? (yes/no/why): no
Are you connected to Wi-Fi but have no internet access? (yes/no/why): no
-----
No rule matched your answers, so no diagnosis can be made.
```

Rules R05, R09 and R06 all need `no_wifi`; after the first "no" the engine found the answer in working memory and skipped those rules without asking again. Rules R07 and R08 share `no_internet` in the same way, so only two questions were needed.

TC-I07 – invalid answer:

```
Is your printer not printing or not responding? (yes/no/why): maybe
Please type yes, no or why:
```

```

Enter your choice: 4

=== Blue screen (stop code) errors ===
(Answer yes or no. Type why to see why a question is asked.)

Are you getting a blue screen (stop code) error? (yes/no/why): yes

Did you add new hardware before the error started? (yes/no/why): no

Does the PC have trouble restarting? (yes/no/why): yes

-----
DIAGNOSIS      : Windows cannot start normally
RECOMMENDATION : Start the PC in Safe Mode to continue troubleshooting.
RULE / SOURCE  : R15 - Microsoft Support: Resolving Blue Screen errors in Windows
-----

Press Enter to return to the menu...

```

Screenshot 6: blue screen consultation where rule R14 is skipped and R15 fires (TC-I04)

```

Enter your choice: 1

=== Sound / audio problems ===
(Answer yes or no. Type why to see why a question is asked.)

Is there no sound coming from your PC? (yes/no/why): yes

Is the wrong (or no) output device selected in Sound settings? (yes/no/why): no

Are your speakers/headphones unplugged, powered off or muted? (yes/no/why): no

Did the problem start after a Windows update? (yes/no/why): no

Does your audio device show a warning in Device Manager? (yes/no/why): no

-----
No rule matched your answers, so no diagnosis can be made.
Please contact a technician or Microsoft Support.
-----

Press Enter to return to the menu...

```

Screenshot 7: sound consultation ending with 'no diagnosis' (TC-I05)

```
Enter your choice: 2

=== Wi-Fi / internet problems ===
(Answer yes or no. Type why to see why a question is asked.)

Are you unable to connect to Wi-Fi? (yes/no/why): no

Are you connected to Wi-Fi but have no internet access? (yes/no/why): no

-----
No rule matched your answers, so no diagnosis can be made.
Please contact a technician or Microsoft Support.
-----

Press Enter to return to the menu...
```

Screenshot 8: Wi-Fi consultation showing each question asked only once (TC-I06)

```
Enter your choice: 3

=== Printer problems ===
(Answer yes or no. Type why to see why a question is asked.)

Is your printer not printing or not responding? (yes/no/why): maybe
Please type yes, no or why: 
```

```
Enter your choice: abc

Invalid choice. Please enter a number from the menu.

Press Enter to return to the menu...
```

Screenshot 9: invalid answer 'maybe' and invalid menu choice 'abc' (TC-I07, TC-I08)

6.4 Summary

40 of 40 automated tests passed and 11 of 11 interactive tests gave the expected output. All 34 rules fire correctly when their conditions are true, the system gives no diagnosis when no rule matches, the more specific rule wins when several rules could apply, questions are never repeated, and invalid input is handled.

7. How to Run the System Locally

7.1 Requirements

SWI-Prolog 9 or later (tested with 10.0.2), free from <https://www.swi-prolog.org/download/stable>. No other software is needed.

7.2 Steps

1. Install SWI-Prolog. On Windows, tick **Add swipl to the system PATH** during installation. Check with `swipl --version`.
2. Download the code: open the GitHub repository, click **Code** → **Download ZIP**, and extract it.

3. Open a terminal in the extracted folder (the folder that contains main.pl). On Windows: open the folder in File Explorer, click the address bar, type cmd and press Enter.
4. Start the system with `swipl main.pl`.
5. Type a menu number and press Enter. Answer the questions with yes, no or why. Choose **9** to see HOW the result was reached and **0** to exit.
6. Optional: run the automated tests with `swipl -g run_all_tests -t halt tests.pl`.

Repository: <https://github.com/Heshan2691/pc-doctor-expert-system.git>

The repository README contains the same instructions, an example session and a troubleshooting table.

7.3 Project files

File	Purpose
main.pl	User interface – start the system with this file
engine.pl	Inference engine and explanation facility
knowledge_base.pl	Rules, questions, categories and sources
tests.pl	40 automated test cases
README.md	Installation and usage instructions
docs/	Rule table, test cases and diagrams

8. References

- [S1] Microsoft (n.d.) *Fix sound or audio problems in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/fix-sound-or-audio-problems-in-windows-73025246-b61c-40fb-671a-2535c7cd56c8> (Accessed: 28 September 2026).
- [S2] Microsoft (n.d.) *Fix Wi-Fi connection issues in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/fix-wi-fi-connection-issues-in-windows-9424a1f7-6a3b-65a6-4d78-7f07eee84d2c> (Accessed: 28 September 2026).
- [S3] Microsoft (n.d.) *Fix printer connection and printing problems in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/fix-printer-connection-and-printing-problems-in-windows-fb830bff-7702-6349-33cd-9443fe987f73> (Accessed: 28 September 2026).
- [S4] Microsoft (n.d.) *Resolving Blue Screen errors in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/topic/60b01860-58f2-be66-7516-5c45a66ae3c6> (Accessed: 28 September 2026).
- [S5] Microsoft (n.d.) *Troubleshooting blank screens in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/hardware/display-graphics/troubleshooting-blank-screens-in-windows> (Accessed: 28 September 2026).
- [S6] Microsoft (n.d.) *Fix Bluetooth problems in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/fix-bluetooth-problems-in-windows-723e092f-03fa-858b-5c80-131ec3fba75c> (Accessed: 28 September 2026).
- [S7] Microsoft (n.d.) *Tips to improve PC performance in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/experience/performance-optimization/tips-to-improve-pc-performance-in-windows> (Accessed: 28 September 2026).
- [S8] Microsoft (n.d.) *Troubleshoot problems updating Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/troubleshoot-problems-updating-windows-188c2b0f-10a7-d72f-65b8-32d177eb136c> (Accessed: 28 September 2026).
- [S9] Microsoft (n.d.) *Free up drive space in Windows*. Microsoft Support. Available at: <https://support.microsoft.com/en-us/windows/experience/storage-filemanagement/free-up-drive-space-in-windows> (Accessed: 28 September 2026).

[S10] SWI-Prolog (n.d.) *SWI-Prolog Reference Manual*. Available at: https://www.swi-prolog.org/pldoc/doc_for?object=manual (Accessed: 28 September 2026).

Annex A – Complete Rule Base (34 rules)

Every rule with its conditions, conclusion and source. Section in source gives the heading on the Microsoft Support page; Source wording quotes the condition stated on the page, or says Listed troubleshooting step when the page lists the step without a specific condition. Rule IDs follow the rule table; the order in which the engine tries the rules is given in Annex B.

Rule	IF (conditions)	THEN (diagnosis → action)	Src	Section in source / Source wording
R01	No sound AND the wrong (or no) output device is selected	Audio output device is misconfigured → Open Settings > System > Sound and select the correct default output device.	S1	Check your speaker output in Windows settings; Set default audio device Listed troubleshooting step
R02	No sound AND speakers/headphones are unplugged, powered off or muted	Audio hardware connection or volume issue → Check the cables, speaker power and volume/mute controls.	S1	Check your audio hardware and volume controls Listed troubleshooting step
R03	No sound AND the problem started after installing Windows updates	Audio driver is incompatible after a Windows update → Roll back the audio driver in Device Manager.	S1	Update or reinstall audio drivers "If you have audio issues after installing updates, try rolling back your audio driver"
R04	No sound AND audio device shows a warning / driver problem in Device Manager	Outdated or corrupt audio driver → Update, or uninstall and reinstall, the audio driver in Device Manager.	S1	Update or reinstall audio drivers Listed troubleshooting step
R05	Cannot connect to Wi-Fi AND Airplane mode is ON	Wireless radios are disabled by Airplane mode → Turn Airplane mode off and make sure Wi-Fi is turned on.	S2	Step 2. Check Airplane Mode Ensure Airplane mode is turned off
R06	Cannot connect to a previously saved Wi-Fi network	Saved Wi-Fi network profile problem → Forget the network, then reconnect and enter the password again.	S2	Step 3. Forget and reconnect to the Wi-Fi Network Listed troubleshooting step
R07	No internet AND other devices on the same network also cannot connect	Router or modem problem (not the PC) → Restart the modem and wireless router (unplug, wait, plug back in).	S2	Step 4. Restart your modem and wireless router; Step 6. Try to connect to the same network on a different device Step 6 helps identify whether the problem is device-specific
R08	No internet AND other devices CAN connect to the same network	Network configuration problem on this PC → Run in an admin Command Prompt: netsh winsock reset, netsh int ip reset, ipconfig /release, ipconfig /renew, ipconfig /flushdns.	S2	Step 7. Run network commands Listed troubleshooting step (after Step 6 shows the issue is device-specific)
R09	Wi-Fi stopped working after a recent update	Network adapter driver problem after a recent update → Uninstall the network adapter driver in Device Manager and restart the PC.	S2	Step 8. Uninstall the network adapter driver and restart Consider if the connection stopped after a recent update
R10	Printer is not printing / not responding	Printer is in a stuck state → Turn off and unplug the printer, wait 30 seconds, then plug it in and turn it on.	S3	Step 1. Unplug and restart your printer Listed troubleshooting step
R11	Printer not detected AND cable is loose or wireless is disconnected	Printer connection problem → Check the USB cable or the printer wireless connection.	S3	Step 2. Check cables or wireless connection Listed troubleshooting step
R12	Print jobs are stuck in the queue	Print spooler problem → Clear and reset the print spooler service.	S3	Step 5. Clear and reset the print spooler "If the previous troubleshooting step is not successful"
R13	Printer shows an "Offline" status	Printer is set to Offline → Change the printer status to "online" in Settings > Bluetooth & devices > Printers & scanners.	S3	Step 6. Change a printer's status to "online" "If your printer displays an 'offline' status"
R14	Blue screen error AND new hardware was added before the error	Newly added hardware is causing the stop error → Remove the new hardware and restart the PC.	S4	Remove any new hardware "If you added new hardware to your PC before the error"
R15	Blue screen error AND PC has trouble restarting	Windows cannot start normally → Start the PC in Safe Mode to continue troubleshooting.	S4	Start your PC in safe mode "If you're having trouble restarting"
R16	Blue screen error AND a device shows a warning (!) in Device Manager	Faulty device driver → Open Device Manager and update the driver of the device marked with (!).	S4	Check the Device Manager Listed troubleshooting step
R17	Blue screen error AND hard drive is almost full	Insufficient free disk space → Free up space on the hard drive (Settings > System > Storage).	S4 / S9	Check for sufficient free space on the hard drive Listed troubleshooting step

Rule	IF (conditions)	THEN (diagnosis → action)	Src	Section in source / Source wording
R18	Blue screen error continues after all the above steps	Persistent system problem → Restore Windows using System Restore or the recovery options.	S4	Restore Windows 11 / Restore Windows 10 "If none of these steps help"
R19	Screen is blank AND an external monitor / cable is used	Display connection problem → Check the cables, try another monitor or cable; on a laptop disconnect external displays.	S5	1. Check Hardware Connections Listed troubleshooting step
R20	Screen is blank BUT the mouse cursor is visible	Windows Explorer has stalled → Press Ctrl+Shift+Esc, find Windows Explorer and select Restart (or run explorer.exe).	S5	4. Restart Windows Explorer If the cursor is visible on the blank screen
R21	Screen is blank AND the problem followed an update	Graphics driver problem after an update → Boot into Safe Mode and roll back the graphics driver in Device Manager.	S5	3. Update or Roll Back Graphics Drivers Roll back if the issue followed an update
R22	Screen is blank AND the problem began recently (not after an update)	A recent system change caused the problem → Perform a System Restore to a restore point before the problem started.	S5	5. Perform a System Restore If the issue began recently
R23	Bluetooth device not connecting AND Bluetooth is turned off	Bluetooth is disabled on the PC → Turn Bluetooth on (also check the physical switch on laptops).	S6	Step 2. Make sure Bluetooth is turned on Listed troubleshooting step
R24	Bluetooth device not connecting AND device is off, uncharged or out of range	Bluetooth device is not ready → Turn on and charge the device, put it in pairing mode and bring it within range.	S6	Step 3. Check your Bluetooth device Device must be powered on, charged and in range
R25	Bluetooth problem started after a recent Windows upgrade or update	Outdated or incompatible Bluetooth driver → Install the latest Bluetooth drivers (Device Manager or Windows Update).	S6	Step 7. Make sure you have the latest drivers After a recent Windows upgrade or updates
R26	Bluetooth icon missing BUT Bluetooth adapter appears in Device Manager	Bluetooth adapter is not loaded properly → Uninstall the Bluetooth adapter in Device Manager and restart the PC.	S6	Step 8. Uninstall the Bluetooth adapter in Device Manager Bluetooth missing but visible in Device Manager
R27	PC is slow AND disk space is low	Low disk space is reducing performance → Free up disk space using Storage Sense or Cleanup recommendations.	S7 / S9	Free up disk space to boost PC performance "Low disk space can reduce system performance"
R28	PC is slow AND it takes a long time to start up	Too many startup apps → Disable unnecessary startup apps (Task Manager > Startup apps).	S7	Disable startup apps to reduce the background activity "Apps that run at startup can slow boot time and increase background usage"
R29	PC is slow AND many apps or browser tabs are open	Too many open apps are using resources → Close unused apps and browser tabs, then restart the PC.	S7	Close unused apps and restart your PC "Having many apps or browser tabs open can slow performance"
R30	PC is slow AND shows suspicious behaviour (pop-ups, unknown programs)	Possible malware infection → Run a full malware scan (Windows Security) and remove any threats.	S7	Run a malware scan to detect and remove threats that slow PC performance "Malware can consume CPU and disk resources, which can reduce system performance"
R31	Windows Update fails AND internet connection is unstable or PC is not plugged in	Update cannot download or install → Plug the device into power and connect to a stable internet connection.	S8	Step 2: Verify that your device is properly plugged in and connected to the Internet "A stable internet connection is crucial for downloading and installing updates."
R32	Windows Update fails AND free disk space is below 20 GB (64-bit) / 16 GB (32-bit)	Not enough free space for the update → Free up drive space (at least 20 GB on 64-bit Windows) and run Windows Update again.	S8 / S9	Step 8. Free up some space so you can run updates "Your device requires at least 16 GB of free space to upgrade a 32-bit OS, or 20 GB for a 64-bit OS."
R33	Windows Update fails AND the date/time on the PC is wrong	Incorrect date and time settings → Correct the date and time in Settings > Time & language.	S8	Step 6. Verify Date and Time settings "Incorrect date and time settings can interfere with Windows Update."
R34	Windows Update fails with an error even though connection, space and date/time are OK	Corrupt update files in the Windows Update cache → Clear the Windows Update cache and run Windows Update again.	S8	Step 5. Clear the Windows Update Cache "Corrupt update files in the cache can cause errors."

Annex B – Decision-Making Code

The complete decision-making logic: the knowledge base (all 34 rules) and the inference engine. The user interface (main.pl) and the tests (tests.pl) are available in the code repository.

B.1 knowledge_base.pl

```
/* =====
PC-Doctor : Windows Troubleshooting Expert System
FILE      : knowledge_base.pl (KNOWLEDGE BASE)
-----
This file contains ONLY knowledge - no reasoning code.
It has four kinds of facts:

1. source(ID, Title, URL)
   - the Microsoft Support pages every rule comes from.

2. category(ID, MenuLabel)
   - the problem areas shown to the user in the menu.

3. question(Fact, Text)
   - the yes/no question the system asks to find out whether
     Fact is true for the user's computer.

4. rule(RuleID, Category, Conditions, Diagnosis, Action, SourceID)
   - IF all Conditions are true
     THEN the problem is Diagnosis and the fix is Action.
   - Conditions is a list of facts, e.g. [no_sound, after_update].
   - Rules inside a category are ordered from MOST SPECIFIC to
     LEAST SPECIFIC, because the inference engine tries them
     from top to bottom and stops at the first rule that fires.

All sources accessed on 28 September 2026.
===== */

:- discontinuous rule/6.

/* -----
1. SOURCES
----- */

source(s1, 'Fix sound or audio problems in Windows',
  'https://support.microsoft.com/en-us/windows/fix-sound-or-audio-problems-in-windows-73025246-b61c-40fb-671a-2535c7cd56c8').
source(s2, 'Fix Wi-Fi connection issues in Windows',
  'https://support.microsoft.com/en-us/windows/fix-wi-fi-connection-issues-in-windows-9424a1f7-6a3b-65a6-4d78-7f07eee84d2c').
source(s3, 'Fix printer connection and printing problems in Windows',
  'https://support.microsoft.com/en-us/windows/fix-printer-connection-and-printing-problems-in-windows-fb830bff-7702-6349-33cd-9443fe987f73').
source(s4, 'Resolving Blue Screen errors in Windows',
  'https://support.microsoft.com/en-us/topic/60b01860-58f2-be66-7516-5c45a66ae3c6').
source(s5, 'Troubleshooting blank screens in Windows',
  'https://support.microsoft.com/en-us/windows/hardware/display-graphics/troubleshooting-blank-screens-in-windows').
source(s6, 'Fix Bluetooth problems in Windows',
  'https://support.microsoft.com/en-us/windows/fix-bluetooth-problems-in-windows-723e092f-03fa-858b-5c80-131ec3fba75c').
source(s7, 'Tips to improve PC performance in Windows',
  'https://support.microsoft.com/en-us/windows/experience/performance-optimization/tips-to-improve-pc-performance-in-windows').
source(s8, 'Troubleshoot problems updating Windows',
  'https://support.microsoft.com/en-us/windows/troubleshoot-problems-updating-windows-188c2b0f-10a7-d72f-65b8-32d177eb136c').
source(s9, 'Free up drive space in Windows',
  'https://support.microsoft.com/en-us/windows/experience/storage-filemanagement/free-up-drive-space-in-windows').

/* -----
2. CATEGORIES (problem areas for the menu)
----- */
```

```

category(sound,          'Sound / audio problems').
category(wifi,           'Wi-Fi / internet problems').
category(printer,        'Printer problems').
category(blue_screen,    'Blue screen (stop code) errors').
category(blank_screen,   'Blank or black screen').
category(bluetooth,      'Bluetooth problems').
category(slow_pc,        'Slow PC performance').
category(windows_update, 'Windows Update problems').

/* -----
3. QUESTIONS (one per fact used in the rules)
----- */

% Sound
question(no_sound,          'Is there no sound coming from your PC?').
question(wrong_output_device, 'Is the wrong (or no) output device selected in Sound settings?').
question(audio_hardware_muted_or_unplugged, 'Are your speakers/headphones unplugged, powered off or muted?').
question(audio_driver_problem, 'Does your audio device show a warning in Device Manager?').

% Shared by several categories
question(after_update,      'Did the problem start after a Windows update?').
question(low_disk_space,    'Is your hard drive almost full?').

% Wi-Fi
question(no_wifi,           'Are you unable to connect to Wi-Fi?').
question(airplane_mode_on,  'Is Airplane mode turned on?').
question(saved_network_fails, 'Does a previously saved Wi-Fi network fail to connect?').
question(no_internet,       'Are you connected to Wi-Fi but have no internet access?').
question(other_devices_fail, 'Do other devices on the same network also have no internet?').
question(other_devices_ok,   'Do other devices on the same network work normally?').

% Printer
question(printer_not_working, 'Is your printer not printing or not responding?').
question(printer_connection_problem, 'Is the printer cable loose or its wireless connection down?').
question(jobs_stuck_in_queue, 'Are print jobs stuck in the print queue?').
question(printer_offline,     'Does the printer show an "Offline" status?').

% Blue screen
question(blue_screen,        'Are you getting a blue screen (stop code) error?').
question(new_hardware_added, 'Did you add new hardware before the error started?').
question(cannot_restart,     'Does the PC have trouble restarting?').
question(device_manager_warning, 'Does any device show a warning (!) in Device Manager?').
question(steps_failed,        'Does the problem continue after trying the previous steps?').

% Blank screen
question(blank_screen,        'Is your screen blank (black)?').
question(external_display,    'Are you using an external monitor or display cable?').
question(cursor_visible,      'Can you see the mouse cursor on the blank screen?').
question(began_recently,      'Did the problem begin recently (not after an update)?').

% Bluetooth
question(bluetooth_problem,   'Is a Bluetooth device not connecting / Bluetooth not working?').
question(bluetooth_off,       'Is Bluetooth turned off on your PC?').
question(bt_device_not_ready, 'Is the Bluetooth device off, uncharged or out of range?').
question(bt_missing_but_in_device_manager, 'Is Bluetooth missing from Settings but shown in Device Manager?').

% Slow PC
question(slow_pc,             'Is your PC running slowly?').
question(slow_startup,        'Does your PC take a long time to start up?').
question(manys_apps_open,     'Do you have many apps or browser tabs open?').
question(suspicious_activity, 'Do you see pop-ups or unknown programs running?').

% Windows Update
question(update_fails,        'Is Windows Update failing?').
question(no_stable_internet,   'Is your internet unstable, or is the PC not plugged into power?').
question(wrong_date_time,      'Is the date or time on your PC wrong?').
question(update_error_persists, 'Does the update still fail although internet, space and date/time are fine?').

/* -----
4. RULES
----- */

```

```

    rule(ID, Category, [Conditions], Diagnosis, Action, Source)
    ----- */

% ----- SOUND (Source S1) -----

% R01 - S1: "Check your speaker output in Windows settings" / "Set default audio device"
rule(r01, sound, [no_sound, wrong_output_device],
    'Audio output device is misconfigured',
    'Open Settings > System > Sound and select the correct default output device.',
    s1).

% R02 - S1: "Check your audio hardware and volume controls"
rule(r02, sound, [no_sound, audio_hardware_muted_or_unplugged],
    'Audio hardware connection or volume issue',
    'Check the cables, speaker power and volume/mute controls.',
    s1).

% R03 - S1: "If you have audio issues after installing updates, try rolling back your audio driver"
rule(r03, sound, [no_sound, after_update],
    'Audio driver is incompatible after a Windows update',
    'Roll back the audio driver in Device Manager.',
    s1).

% R04 - S1: "Update or reinstall audio drivers"
rule(r04, sound, [no_sound, audio_driver_problem],
    'Outdated or corrupt audio driver',
    'Update, or uninstall and reinstall, the audio driver in Device Manager.',
    s1).

% ----- WI-FI (Source S2) -----

% R05 - S2: "Step 2. Check Airplane Mode"
rule(r05, wifi, [no_wifi, airplane_mode_on],
    'Wireless radios are disabled by Airplane mode',
    'Turn Airplane mode off and make sure Wi-Fi is turned on.',
    s2).

% R09 - S2: "Step 8. Uninstall the network adapter driver and restart"
% (placed before R06 because it is more specific)
rule(r09, wifi, [no_wifi, after_update],
    'Network adapter driver problem after a recent update',
    'Uninstall the network adapter driver in Device Manager and restart the PC.',
    s2).

% R06 - S2: "Step 3. Forget and reconnect to the Wi-Fi Network"
rule(r06, wifi, [no_wifi, saved_network_fails],
    'Saved Wi-Fi network profile problem',
    'Forget the network, then reconnect and enter the password again.',
    s2).

% R07 - S2: "Step 4. Restart your modem and wireless router"
% + "Step 6. Try to connect to the same network on a different device"
rule(r07, wifi, [no_internet, other_devices_fail],
    'Router or modem problem (not the PC)',
    'Restart the modem and wireless router (unplug, wait, plug back in).',
    s2).

% R08 - S2: "Step 7. Run network commands"
rule(r08, wifi, [no_internet, other_devices_ok],
    'Network configuration problem on this PC',
    'Run in an admin Command Prompt: netsh winsock reset, netsh int ip reset, ipconfig /release, ipconfig /renew, ipconfig /flushdns.',
    s2).

% ----- PRINTER (Source S3) -----

% R13 - S3: "Step 6. Change a printer's status to online"
rule(r13, printer, [printer_not_working, printer_offline],
    'Printer is set to Offline',
    'Change the printer status to "online" in Settings > Bluetooth & devices > Printers & scanners.',
    s3).

% R12 - S3: "Step 5. Clear and reset the print spooler"
rule(r12, printer, [printer_not_working, jobs_stuck_in_queue],
    'Print spooler problem',
    'Clear and reset the print spooler service.',
    s3).

```

```

s3).

% R11 - S3: "Step 2. Check cables or wireless connection"
rule(r11, printer, [printer_not_working, printer_connection_problem],
    'Printer connection problem',
    'Check the USB cable or the printer wireless connection.',
    s3).

% R10 - S3: "Step 1. Unplug and restart your printer"
% (general rule - placed LAST in this category)
rule(r10, printer, [printer_not_working],
    'Printer is in a stuck state',
    'Turn off and unplug the printer, wait 30 seconds, then plug it in and turn it on.',
    s3).

% ----- BLUE SCREEN (Source S4, S9) -----

% R14 - S4: "Remove any new hardware"
rule(r14, blue_screen, [blue_screen, new_hardware_added],
    'Newly added hardware is causing the stop error',
    'Remove the new hardware and restart the PC.',
    s4).

% R15 - S4: "Start your PC in safe mode"
rule(r15, blue_screen, [blue_screen, cannot_restart],
    'Windows cannot start normally',
    'Start the PC in Safe Mode to continue troubleshooting.',
    s4).

% R16 - S4: "Check the Device Manager"
rule(r16, blue_screen, [blue_screen, device_manager_warning],
    'Faulty device driver',
    'Open Device Manager and update the driver of the device marked with (!).',
    s4).

% R17 - S4: "Check for sufficient free space on the hard drive" (see also S9)
rule(r17, blue_screen, [blue_screen, low_disk_space],
    'Insufficient free disk space',
    'Free up space on the hard drive (Settings > System > Storage).',
    s4).

% R18 - S4: "Restore Windows" - "If none of these steps help"
rule(r18, blue_screen, [blue_screen, steps_failed],
    'Persistent system problem',
    'Restore Windows using System Restore or the recovery options.',
    s4).

% ----- BLANK SCREEN (Source S5) -----

% R20 - S5: "4. Restart Windows Explorer" (if the cursor is visible)
rule(r20, blank_screen, [blank_screen, cursor_visible],
    'Windows Explorer has stalled',
    'Press Ctrl+Shift+Esc, find Windows Explorer and select Restart (or run explorer.exe).',
    s5).

% R19 - S5: "1. Check Hardware Connections"
rule(r19, blank_screen, [blank_screen, external_display],
    'Display connection problem',
    'Check the cables, try another monitor or cable; on a laptop disconnect external displays.',
    s5).

% R21 - S5: "3. Update or Roll Back Graphics Drivers" (roll back if after an update)
rule(r21, blank_screen, [blank_screen, after_update],
    'Graphics driver problem after an update',
    'Boot into Safe Mode and roll back the graphics driver in Device Manager.',
    s5).

% R22 - S5: "5. Perform a System Restore" (if the issue began recently)
rule(r22, blank_screen, [blank_screen, began_recently],
    'A recent system change caused the problem',
    'Perform a System Restore to a restore point before the problem started.',
    s5).

% ----- BLUETOOTH (Source S6) -----

% R23 - S6: "Step 2. Make sure Bluetooth is turned on"

```

```

rule(r23, bluetooth, [bluetooth_problem, bluetooth_off],
    'Bluetooth is disabled on the PC',
    'Turn Bluetooth on (also check the physical switch on laptops).',
    s6).

% R24 - S6: "Step 3. Check your Bluetooth device"
rule(r24, bluetooth, [bluetooth_problem, bt_device_not_ready],
    'Bluetooth device is not ready',
    'Turn on and charge the device, put it in pairing mode and bring it within range.',
    s6).

% R26 - S6: "Step 8. Uninstall the Bluetooth adapter in Device Manager"
rule(r26, bluetooth, [bluetooth_problem, bt_missing_but_in_device_manager],
    'Bluetooth adapter is not loaded properly',
    'Uninstall the Bluetooth adapter in Device Manager and restart the PC.',
    s6).

% R25 - S6: "Step 7. Make sure you have the latest drivers" (after an upgrade/update)
rule(r25, bluetooth, [bluetooth_problem, after_update],
    'Outdated or incompatible Bluetooth driver',
    'Install the latest Bluetooth drivers (Device Manager or Windows Update).',
    s6).

% ----- SLOW PC (Source S7, S9) -----

% R30 - S7: "Run a malware scan to detect and remove threats that slow PC performance"
rule(r30, slow_pc, [slow_pc, suspicious_activity],
    'Possible malware infection',
    'Run a full malware scan (Windows Security) and remove any threats.',
    s7).

% R27 - S7: "Free up disk space to boost PC performance" (see also S9)
rule(r27, slow_pc, [slow_pc, low_disk_space],
    'Low disk space is reducing performance',
    'Free up disk space using Storage Sense or Cleanup recommendations.',
    s7).

% R28 - S7: "Disable startup apps to reduce the background activity"
rule(r28, slow_pc, [slow_pc, slow_startup],
    'Too many startup apps',
    'Disable unnecessary startup apps (Task Manager > Startup apps).',
    s7).

% R29 - S7: "Close unused apps and restart your PC"
rule(r29, slow_pc, [slow_pc, many_apps_open],
    'Too many open apps are using resources',
    'Close unused apps and browser tabs, then restart the PC.',
    s7).

% ----- WINDOWS UPDATE (Source S8, S9) -----

% R31 - S8: "Step 2: Verify that your device is properly plugged in and connected to the Internet"
rule(r31, windows_update, [update_fails, no_stable_internet],
    'Update cannot download or install',
    'Plug the device into power and connect to a stable internet connection.',
    s8).

% R32 - S8: "Step 8. Free up some space so you can run updates" (16 GB / 20 GB)
rule(r32, windows_update, [update_fails, low_disk_space],
    'Not enough free space for the update',
    'Free up drive space (at least 20 GB on 64-bit Windows) and run Windows Update again.',
    s8).

% R33 - S8: "Step 6. Verify Date and Time settings"
rule(r33, windows_update, [update_fails, wrong_date_time],
    'Incorrect date and time settings',
    'Correct the date and time in Settings > Time & language.',
    s8).

% R34 - S8: "Step 5. Clear the Windows Update Cache"
rule(r34, windows_update, [update_fails, update_error_persists],
    'Corrupt update files in the Windows Update cache',
    'Clear the Windows Update cache and run Windows Update again.',
    s8).

```

```

/* -----
5. KNOWLEDGE BASE CHECK (helper for the developer)
   Run ?- kb_check. to make sure every condition has a question,
   every rule has a valid category and source.
----- */

kb_check :-
    findall(R, rule(R,_,_,_,_), Rules),
    length(Rules, N),
    format("Rules loaded: ~w~n", [N]),
    forall(( rule(R,_,Conds,_,_), member(C, Conds), \+ question(C,_) ),
        format(" WARNING: ~w uses ~w but it has no question~n", [R, C])),
    forall(( rule(R,Cat,_,_,_), \+ category(Cat,_) ),
        format(" WARNING: ~w has unknown category ~w~n", [R, Cat])),
    forall(( rule(R,_,_,_,S), \+ source(S,_) ),
        format(" WARNING: ~w has unknown source ~w~n", [R, S])),
    format("Check finished.~n").

```

B.2 engine.pl

```

/* =====
PC-Doctor : Windows Troubleshooting Expert System
FILE      : engine.pl (INFERENCE ENGINE + EXPLANATION FACILITY)
-----

How it works (BACKWARD CHAINING):

1. The user picks a problem category (e.g. printer).
2. The engine takes the rules of that category from top to bottom.
3. For each rule it tries to prove every condition in its IF list:
   - if the answer is already in WORKING MEMORY (known/2) it is reused,
   - otherwise the user is ASKED the question and the answer is stored.
4. The FIRST rule whose conditions are all "yes" FIRES and gives the
   diagnosis. If no rule fires, the system says it cannot diagnose.

Explanation facility:
- WHY : while a question is shown the user can type "why" to see which
        rule the engine is currently testing.
- HOW : after a diagnosis, how/0 shows which rule fired, the facts that
        made it true, and the Microsoft source of the rule.
===== */

:- ensure_loaded(knowledge_base).

:- dynamic known/2.      % known(Fact, yes/no) - working memory
:- dynamic fired/1.      % fired(RuleID)       - rule that gave the result
:- dynamic batch_mode/0. % when set, questions are not asked (used for tests)

/* -----
1. WORKING MEMORY
----- */

reset :-
    retractall(known(_, _)),
    retractall(fired(_)).

/* -----
2. INFERENCE - backward chaining
----- */

% diagnose(+Category, -RuleID)
% Find the first rule of Category whose conditions are all true.
diagnose(Category, RuleID) :-
    rule(RuleID, Category, Conditions, _, _),
    all_true(Conditions, RuleID),
    !,
    assertz(fired(RuleID)).

% all_true(+Conditions, +RuleID) - every condition in the list is true
all_true([], _).
all_true([Fact | Rest], RuleID) :-
    is_true(Fact, RuleID),
    all_true(Rest, RuleID).

```

```

% is_true(+Fact, +RuleID)
is_true(Fact, _) :-                % already known to be true
    known(Fact, yes), !.
is_true(Fact, _) :-                % already known to be false
    known(Fact, no), !, fail.
is_true(Fact, RuleID) :-           % unknown -> ask the user and remember
    ask(Fact, RuleID, Answer),
    assertz(known(Fact, Answer)),
    Answer == yes.

/* -----
3. ASKING QUESTIONS (user dialogue)
----- */

% In batch (test) mode every unknown fact is taken as "no".
ask(_, _, no) :-
    batch_mode, !.
ask(Fact, RuleID, Answer) :-
    question(Fact, Text),
    format("~n ~w (yes/no/why): ", [Text]),
    read_answer(Reply),
    ( Reply == why
    -> explain_why(RuleID),
        ask(Fact, RuleID, Answer)
    ; Answer = Reply
    ).

% read_answer(-Reply) : Reply is yes, no or why
read_answer(Reply) :-
    read_line_to_string(user_input, Line),
    ( Line == end_of_file
    -> Reply = no
    ; normalize_space(atom(A), Line),
        downcase_atom(A, Word),
        ( answer_word(Word, Reply)
        -> true
        ; format(" Please type yes, no or why: "),
            read_answer(Reply)
        )
    ).

answer_word(yes, yes).  answer_word(y, yes).
answer_word(no, no).   answer_word(n, no).
answer_word(why, why). answer_word(w, why).  answer_word('?', why).

/* -----
4. EXPLANATION FACILITY
----- */

% WHY - shown during questioning
explain_why(RuleID) :-
    rule(RuleID, _, Conditions, Diagnosis, _, Source),
    source(Source, Title, _),
    upcase_atom(RuleID, R),
    format("~n [WHY] I am testing rule ~w:~n", [R]),
    format("      IF~n"),
    forall(member(C, Conditions), print_condition(C)),
    format("      THEN ~w~n", [Diagnosis]),
    format("      (Source: ~w)~n", [Title]).

print_condition(C) :-
    question(C, Text),
    ( known(C, V) -> Status = V ; Status = 'not asked yet' ),
    format("      - ~w [~w]~n", [Text, Status]).

% HOW - shown after a diagnosis
how :-
    fired(RuleID), !,
    rule(RuleID, _, Conditions, Diagnosis, _, Source),
    source(Source, Title, URL),
    upcase_atom(RuleID, R),
    upcase_atom(Source, S),
    format("~n [HOW] The conclusion was reached by rule ~w:~n", [R]),
    format("      IF~n"),

```

```

forall(member(C, Conditions), print_condition(C)),
format("          THEN ~w~n", [Diagnosis]),
format("~n          Source ~w: ~w~n          ~w~n", [S, Title, URL]),
rejected_rules.
how :-
    format("~n No diagnosis has been made yet.~n").

% rejected_rules - rules that were ruled out by a "no" answer
rejected_rules :-
    fired(Fired),
    rule(Fired, Category, _, _, _),
    findall(R, ( rule(R, Category, Conds, _, _),
                R \== Fired,
                member(C, Conds), known(C, no) ),
            Rs0),
    sort(Rs0, Rs),
    (   Rs == []
    -> true
    ;   maplist(upcase_atom, Rs, Up),
        atomic_list_concat(Up, ', ', List),
        format("~n          Rules ruled out by your 'no' answers: ~w~n", [List])
    ).

/* -----
5. RUNNING A CONSULTATION
----- */

% consultation(+Category) - full interactive session for one category
consultation(Category) :-
    reset,
    category(Category, Label),
    format("~n=== ~w ===~n", [Label]),
    format(" (Answer yes or no. Type why to see why a question is asked.)~n"),
    (   diagnose(Category, RuleID)
    -> show_result(RuleID)
    ;   no_result
    ).

show_result(RuleID) :-
    rule(RuleID, _, _, Diagnosis, Action, Source),
    source(Source, Title, _),
    upcase_atom(RuleID, R),
    format("~n -----~n"),
    format("  DIAGNOSIS      : ~w~n", [Diagnosis]),
    format("  RECOMMENDATION : ~w~n", [Action]),
    format("  RULE / SOURCE  : ~w - Microsoft Support: ~w~n", [R, Title]),
    format(" -----~n").

no_result :-
    format("~n -----~n"),
    format(" No rule matched your answers, so no diagnosis can be made.~n"),
    format(" Please contact a technician or Microsoft Support.~n"),
    format(" -----~n").

/* -----
6. NON-INTERACTIVE MODE (for test cases)
run_test(+Category, +YesFacts, -Result)
    YesFacts = facts the "user" answers yes to; all others are no.
    Result   = the RuleID that fired, or none.
    Example: ?- run_test(printer, [printer_not_working, printer_offline], R).
              R = r13.
----- */

run_test(Category, YesFacts, Result) :-
    reset,
    assertz(batch_mode),
    forall(member(F, YesFacts), assertz(known(F, yes))),
    (   diagnose(Category, R) -> Result = R ; Result = none ),
    retractall(batch_mode).

```

Annex C – Automated Test Results (40 tests)

Output of `swipl -g run_all_tests -t halt tests.pl`: 40 of 40 passed.

Test	Category	Purpose	User answers	Expected / Actual	Result
T01	Sound / audio problems	Rule R01 fires when its conditions are true	YES: Is there no sound coming from your PC? YES: Is the wrong (or no) output device selected in Sound settings?	R01: Audio output device is misconfigured	PASS
T02	Sound / audio problems	Rule R02 fires when its conditions are true	YES: Is there no sound coming from your PC? YES: Are your speakers/headphones unplugged, powered off or muted?	R02: Audio hardware connection or volume issue	PASS
T03	Sound / audio problems	Rule R03 fires when its conditions are true	YES: Is there no sound coming from your PC? YES: Did the problem start after a Windows update?	R03: Audio driver is incompatible after a Windows update	PASS
T04	Sound / audio problems	Rule R04 fires when its conditions are true	YES: Is there no sound coming from your PC? YES: Does your audio device show a warning in Device Manager?	R04: Outdated or corrupt audio driver	PASS
T05	Wi-Fi / internet problems	Rule R05 fires when its conditions are true	YES: Are you unable to connect to Wi-Fi? YES: Is Airplane mode turned on?	R05: Wireless radios are disabled by Airplane mode	PASS
T06	Wi-Fi / internet problems	Rule R06 fires when its conditions are true	YES: Are you unable to connect to Wi-Fi? YES: Does a previously saved Wi-Fi network fail to connect?	R06: Saved Wi-Fi network profile problem	PASS
T07	Wi-Fi / internet problems	Rule R07 fires when its conditions are true	YES: Are you connected to Wi-Fi but have no internet access? YES: Do other devices on the same network also have no internet?	R07: Router or modem problem (not the PC)	PASS
T08	Wi-Fi / internet problems	Rule R08 fires when its conditions are true	YES: Are you connected to Wi-Fi but have no internet access? YES: Do other devices on the same network work normally?	R08: Network configuration problem on this PC	PASS
T09	Wi-Fi / internet problems	Rule R09 fires when its conditions are true	YES: Are you unable to connect to Wi-Fi? YES: Did the problem start after a Windows update?	R09: Network adapter driver problem after a recent update	PASS
T10	Printer problems	Rule R10 fires when its conditions are true	YES: Is your printer not printing or not responding?	R10: Printer is in a stuck state	PASS
T11	Printer problems	Rule R11 fires when its conditions are true	YES: Is your printer not printing or not responding? YES: Is the printer cable loose or its wireless connection down?	R11: Printer connection problem	PASS
T12	Printer problems	Rule R12 fires when its conditions are true	YES: Is your printer not printing or not responding? YES: Are print jobs stuck in the print queue?	R12: Print spooler problem	PASS
T13	Printer problems	Rule R13 fires when its conditions are true	YES: Is your printer not printing or not responding? YES: Does the printer show an "Offline" status?	R13: Printer is set to Offline	PASS
T14	Blue screen (stop code) errors	Rule R14 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Did you add new hardware before the error started?	R14: Newly added hardware is causing the stop error	PASS
T15	Blue screen (stop code) errors	Rule R15 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Does the PC have trouble restarting?	R15: Windows cannot start normally	PASS
T16	Blue screen (stop code) errors	Rule R16 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Does any device show a warning (!) in Device Manager?	R16: Faulty device driver	PASS
T17	Blue screen (stop code) errors	Rule R17 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Is your hard drive almost full?	R17: Insufficient free disk space	PASS
T18	Blue screen (stop code) errors	Rule R18 fires when its conditions are true	YES: Are you getting a blue screen (stop code) error? YES: Does the problem continue after trying the previous steps?	R18: Persistent system problem	PASS
T19	Blank or black screen	Rule R19 fires when its conditions are true	YES: Is your screen blank (black)? YES: Are you using an external monitor or display cable?	R19: Display connection problem	PASS
T20	Blank or black screen	Rule R20 fires when its conditions are true	YES: Is your screen blank (black)? YES: Can you see the mouse cursor on the blank screen?	R20: Windows Explorer has stalled	PASS
T21	Blank or black screen	Rule R21 fires when its conditions are true	YES: Is your screen blank (black)? YES: Did the problem start after a Windows update?	R21: Graphics driver problem after an update	PASS
T22	Blank or black	Rule R22 fires when its	YES: Is your screen blank (black)?	R22: A recent system	PASS

Test	Category	Purpose	User answers	Expected / Actual	Result
	screen	conditions are true	YES: Did the problem begin recently (not after an update)?	change caused the problem	
T23	Bluetooth problems	Rule R23 fires when its conditions are true	YES: Is a Bluetooth device not connecting / Bluetooth not working? YES: Is Bluetooth turned off on your PC?	R23: Bluetooth is disabled on the PC	PASS
T24	Bluetooth problems	Rule R24 fires when its conditions are true	YES: Is a Bluetooth device not connecting / Bluetooth not working? YES: Is the Bluetooth device off, uncharged or out of range?	R24: Bluetooth device is not ready	PASS
T25	Bluetooth problems	Rule R25 fires when its conditions are true	YES: Is a Bluetooth device not connecting / Bluetooth not working? YES: Did the problem start after a Windows update?	R25: Outdated or incompatible Bluetooth driver	PASS
T26	Bluetooth problems	Rule R26 fires when its conditions are true	YES: Is a Bluetooth device not connecting / Bluetooth not working? YES: Is Bluetooth missing from Settings but shown in Device Manager?	R26: Bluetooth adapter is not loaded properly	PASS
T27	Slow PC performance	Rule R27 fires when its conditions are true	YES: Is your PC running slowly? YES: Is your hard drive almost full?	R27: Low disk space is reducing performance	PASS
T28	Slow PC performance	Rule R28 fires when its conditions are true	YES: Is your PC running slowly? YES: Does your PC take a long time to start up?	R28: Too many startup apps	PASS
T29	Slow PC performance	Rule R29 fires when its conditions are true	YES: Is your PC running slowly? YES: Do you have many apps or browser tabs open?	R29: Too many open apps are using resources	PASS
T30	Slow PC performance	Rule R30 fires when its conditions are true	YES: Is your PC running slowly? YES: Do you see pop-ups or unknown programs running?	R30: Possible malware infection	PASS
T31	Windows Update problems	Rule R31 fires when its conditions are true	YES: Is Windows Update failing? YES: Is your internet unstable, or is the PC not plugged into power?	R31: Update cannot download or install	PASS
T32	Windows Update problems	Rule R32 fires when its conditions are true	YES: Is Windows Update failing? YES: Is your hard drive almost full?	R32: Not enough free space for the update	PASS
T33	Windows Update problems	Rule R33 fires when its conditions are true	YES: Is Windows Update failing? YES: Is the date or time on your PC wrong?	R33: Incorrect date and time settings	PASS
T34	Windows Update problems	Rule R34 fires when its conditions are true	YES: Is Windows Update failing? YES: Does the update still fail although internet, space and date/time are fine?	R34: Corrupt update files in the Windows Update cache	PASS
T35	Printer problems	Main symptom answered NO - system must not give a diagnosis	(all questions answered NO)	No diagnosis	PASS
T36	Sound / audio problems	A detail is YES but the main symptom is NO - no diagnosis	YES: Is the wrong (or no) output device selected in Sound settings?	No diagnosis	PASS
T37	Sound / audio problems	Main symptom YES but no detail matches - no diagnosis	YES: Is there no sound coming from your PC?	No diagnosis	PASS
T38	Printer problems	Several details YES - most specific rule (first in order) must win	YES: Is your printer not printing or not responding? YES: Does the printer show an "Offline" status? YES: Are print jobs stuck in the print queue?	R13: Printer is set to Offline	PASS
T39	Slow PC performance	Several details YES - most specific rule (first in order) must win	YES: Is your PC running slowly? YES: Do you see pop-ups or unknown programs running? YES: Is your hard drive almost full?	R30: Possible malware infection	PASS
T40	Wi-Fi / internet problems	Several details YES - most specific rule (first in order) must win	YES: Are you unable to connect to Wi-Fi? YES: Is Airplane mode turned on? YES: Did the problem start after a Windows update?	R05: Wireless radios are disabled by Airplane mode	PASS